

Designing a Modern Payment Gateway

1. Problem Statement

You are a Software Architect at a Fintech startup. Your platform needs to support multiple mobile financial services (MFS) for checkouts. To keep the code maintainable, you must ensure that every payment provider (like bKash or Nagad) follows a strict **PaymentProcessor** contract. However, some functionality (like logging) is the same for everyone, and some data (like exchange rates) belongs to the system as a whole, not a specific provider.

2. Task Definitions

Task A: The Interface Contract

Create an interface named `PaymentProcessor`.

1. **Abstract Method:** Define a method `void processPayment(double amount)`. Every bank/MFS must implement their own logic for this.
2. **Default Method:** Add a method default `void logTransaction(String status)`.
 - *Implementation:* Simply print: "Logging Transaction: [status]".
 - *Purpose:* This provides a shared logging mechanism so developers don't have to rewrite it for every provider.

Task B: Implementation

Create two concrete classes:

1. **BkashProcessor:** Implements `PaymentProcessor`. Print "Processing bKash payment of [amount] BDT".
2. **NagadProcessor:** Implements `PaymentProcessor`. Print "Processing Nagad payment of [amount] BDT".

Task C: The Static Challenge

1. Add a static method to the `PaymentProcessor` interface:

```
static double getExchangeRate() { return 110.50; }
```

2. In your Main class, try to call this method using an **instance** of a class:

```
PaymentProcessor p = new BkashProcessor();  
System.out.println(p.getExchangeRate());
```

3. Deliverables (Questions to Answer)

Q1. Method Inheritance: Does the BkashProcessor class *have* to override the logTransaction() method? Why or why not?

Q2. The Static Failure: When you tried to call p.getExchangeRate(), what error did the compiler give you? Explain why Java prevents calling interface static methods through an instance, unlike class static methods.

Q3. Correct Usage: What is the correct way to access the getExchangeRate() method? Provide the specific line of code.

4. Sample Code Structure for Reference

```
public class PaymentSystem {  
    public static void main(String[] args) {  
        // 1. Using Interface Reference  
        PaymentProcessor bKash = new BkashProcessor();  
        bKash.processPayment(5000);  
        bKash.logTransaction("SUCCESS");  
        // 2. Testing the Static Method  
        // System.out.println(bKash.getExchangeRate());  
        // TODO: Access the static method correctly here  
    }  
}
```